Doris Ying-Ying Cheng

in linkedin.com/in/yingying-cheng github.com/Yingyingcheng

Portfolio doris82717@gmail.com 628-306-4080

SUMMARY

Full-stack Software Engineer with 5+ years of experience building real-time **multiplayer systems**, **web/TV applications**, and **scalable backend services**. Expert in React, TypeScript, Node.js, **automated CI/CD pipelines, and AI-driven development.**

WORK EXPERIENCE

Software Engineer | Party Monster Games *July 2026 – Present*

- Partnered with the Founder to turn specs into production architecture for Doodle Party and Emoji Party, leading code reviews and version control workflows; overhauled the TV Hub Menu with prefetching and caching to cut load time.
- Fixed critical Fire OS rendering lag by utilizing an **AI-driven workflow (Cursor/Claude)** to troubleshoot event handling in Vega, engineering a **50ms stroke batching queue** that reduced backlogs by **80%–95%** (from 400 to ~20 msgs/sec).
- Architected scalable **REST APIs** and game features, leveraging **RevenueCat/Amplitude data analytics** and AI collaboration across **rapid prototyping** and **automated testing** to accelerate deployment and drive user growth.

Software Engineer Intern | Party Monster Games *Jan 2026 – July 2026*

- Built real-time **WebSocket game-state synchronization** with Colyseus across TV, mobile, and server for Trivia Party and WordRush Party — top-10 ranked Fire TV apps with 10K+ monthly downloads.
- Led end-to-end **web development** of a new game mode (**PostgreSQL** schema, **Node.js/TypeScript** backend, **React frontend UX/UI**), driving **30% MoM MRR growth** in its first month.

Member of Technical Staff | Taipei City Government *Sep 2019 – July 2024*

- Owned the full **development life cycle** of a Python geospatial remote-sensing pipeline, **designing hybrid SQL/NoSQL database models** that eliminated legacy failures and **cut calculation time by 12%** for 1,000+ daily users.

Graduate Research Assistant (Machine Learning) | National Taiwan University *July 2017 – June 2019*

- Engineered a custom **machine learning Python data pipeline** for hyperspectral **image classification** using dictionary-based sparse representation. Published a peer-reviewed conference paper [Link1][Link2].

EDUCATION

San Francisco Bay University *2024 – May 2026*

M.S. in Computer Science, ***Specialization in Web Programming and Data Modeling*** – GPA: 3.9/4.0

National Taiwan University *2017 – 2019*

M.S. in Civil Engineering, ***Specialization in Image Processing*** – GPA CS-related Class: 4.0/4.0

- Thesis: Hyperspectral Image Classification via Integration of Joint Sparse Representation with Spectral and Spatial Information [Link]

PROJECTS

Y2k-flying-bananas (3D AI Playground) – [Live Demo][GitHub] *Jan 2026 – Feb 2026*

- Developed a Windows 98–styled **3D web application** using React Three Fiber, featuring a **multilingual GPT chatbot** and playable mini-games; optimized 3D rendering pipelines and **SEO** to achieve top-tier Core Web Vitals performance.

Foodie Diary (AI Calories tracking app) – [Live Demo][GitHub] *July 2025 – Feb 2026*

- Built a **React/TypeScript** full-stack meal tracker leveraging the **Gemini API** for advanced multimodal image/text analysis, automating precise calorie estimation and structured macro extraction on a serverless backend.

AI Newsreader & Journalist – [Summary][GitHub] *Aug 2025 – Dec 2025*

- Engineered robust **CI/CD pipelines (GitHub Actions, Cloud Build)** to containerize and deploy HeyGen AI video microservices onto **Google Kubernetes Engine (GKE)**, integrating HPA for seamless high availability.

SKILLS & COURSES

Languages: Python, **JavaScript (ES6+)**, **TypeScript**, **SQL**, **HTML/CSS**; Familiar: Java
Frameworks & Tools: React, Next.js, Node.js, Express, Tailwind CSS, Three.js (R3F), Colyseus, REST APIs, WebSockets
Databases & DevOps: PostgreSQL, MySQL, MongoDB, Docker, Kubernetes (GKE), GCP, AWS (Lambda), GitHub Actions, CI/CD
Core Skills & Coursework: Automated Testing, Cloud Infrastructure, Data Modeling, Advanced Web Programming